CAPSTONE PROJECT - EXECUTIVE PRESENTATION

Fraud Detection Using Machine Learning

John Paul Aguinaldo
Post-Graduate Diploma in AI and Machine Learning
March 2026

BUSINESS PROBLEM

Smarter fraud detection is needed to mitigate financial loss, manage operational efficiency, and improve customer experience.

PROBLEM CONTEXT

Credit Card Fraud

- Direct financial losses
- Manual investigation costs
- Customer friction and dissatisfaction
- Reputational and regulatory risk

OPPORTUNITY

Machine Learning

- Early fraud detection
- Reduced false alarms
- Scalable decision-making

SUCCESS MEASURES

KPIs

- Fraud detection rate
- False positive rate
- Operational efficiency

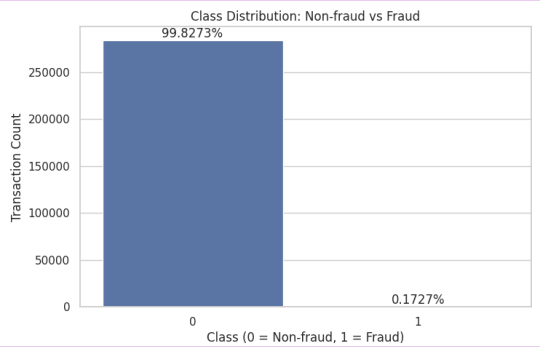
DATA AND INSIGHTS

Credit card fraud is rare but exhibits detectable behavioral patterns.

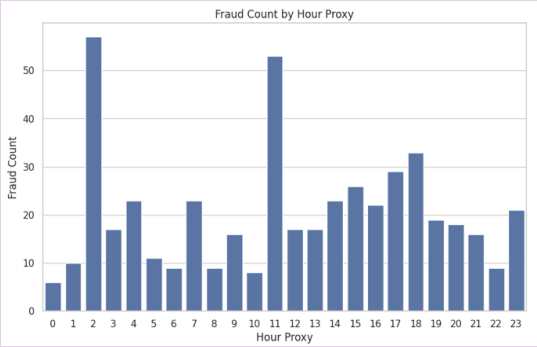
DATASET
Credit Card Fraud Detection

~285K	0.17%	0	1,081
TRANSACTIONS	FRAUD RATE	MISSING DATA	DUPLICATE TRANSACTIONS

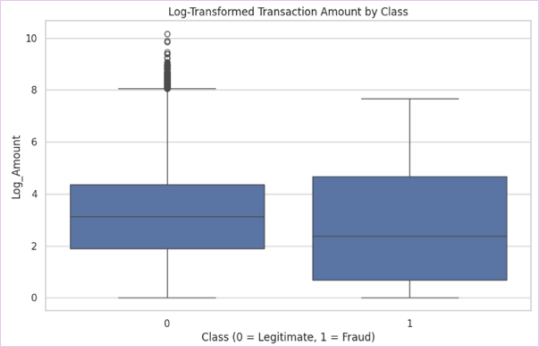
Fraud is **highly imbalanced** and **difficult to detect.**



Fraud risk is higher **at night (12:00 a.m. to 5:00 a.m.).**



Fraud occurs **across transaction sizes**.



MACHINE LEARNING SOLUTION

A structured machine learning (ML) approach was used to identify high-risk transactions and improve fraud detection accuracy.

Solution Design Principles

- Prioritize **fraud detection**
- Balance with **operational efficiency**

HOW IT WORKS
Machine Learning

INPUT
Historical data

→

PROCESS
Learn fraud behavior based on:
- Timing of transactions
- Transaction characteristics
- Hidden patterns in the data

→

OUTPUT
Fraud or Legitimate?

WHAT WAS DONE
Modeling Pipeline

PREPARATION
Cleaning and preparation of transaction data

→

IMPLEMENTATION
Tested multiple analytical approaches

→

MODEL SELECTION
Selected approach with best balance between
- Fraud detection
- Operational efficiency

MODEL PERFORMANCE

The final model produces highly reliable fraud alerts while capturing most fraudulent transactions.

Final Model Performance

- Detects **majority** of fraudulent transactions
- Generates **high-confidence alerts**
- Flags **few false alarms**

Performance Metric	Interpretation	What This Means
Precision **0.96**	**Very few false alarms** 96% of flagged transactions are truly fraudulent.	▪ Fraud teams can focus on higher quality cases. ▪ Customers experience fewer unnecessary disruptions.
Recall **0.79**	**Detects a majority of fraud cases** 79% of fraud is detected.	▪ Strong fraud protection. ▪ BUT some fraud may still be missed.

BUSINESS IMPACT & ROI

The model reduces costs and improves efficiency while maintaining strong fraud protection.

MACHINE LEARNING MODEL

Value Proposition

- **Reduced fraud losses:** More fraud cases detected early
- **Lower operational costs:** Fewer false alarms → less manual review
- **Better customer experience:** Fewer unnecessary transaction disruptions

ML MODEL INTEGRATION

Efficiency Gains

- More efficient fraud investigation process
- Scalable fraud detection capability
- Better resource allocation

RISKS AND LIMITATIONS

Key risks must be actively managed to protect ROI and ensure fairness.

Performance Risks

- Missed fraud cases (false negatives)
- Uneven detection across time segments

Fairness & Governance Risks

- Variability in subgroup performance
- Limited ability to fully assess fairness

Interpretability Risks

Model is technically explainable but not directly actionable due to anonymized features

IMPLICATION

Model cannot operate independently without safeguards.

STRATEGY & RECOMMENDATIONS

Adopt a multi-layered strategy to maximize the model's value and manage fraud risk effectively.

Fraud Detection Strategy

- Deploy model as primary fraud detection layer.
- Adjust thresholds based on risk appetite.

Decision Strategy

Implement a tiered decision framework:

- High-risk → Block
- Medium-risk → Review
- Low-risk → Approve

Risk Management Strategy

Implement continuous monitoring of key metrics:

- Model Performance: Recall, Precision, PR-AUC
- Fairness: Subgroup Performance

Governance & Ethical AI

- Maintain audit logs and model transparency
- Monitor fairness and bias
- Enable human oversight for critical decisions